INVESTMENTS

Ninth Canadian Edition · Bodie, Kane, Marcus

CHAPTER 10

Arbitrage Pricing Theory and Multifactor Models of Risk and Return

Verified against source text — learning objectives, formulas, and worked examples checked directly against the chapter.

Study Guide · Prepared for Personal Use

LO1 · Multifactor Models: An Overview

The single-index model (Chapter 8) summarizes all systematic risk in one macro factor. In reality, distinct economic forces — the business cycle, interest rates, inflation, energy prices — can affect securities differently, and two stocks with the same market beta might respond very differently to, say, an interest-rate shock. **Multifactor models** extend the single-index approach by allowing several distinct systematic factors, each with its own risk premium and each security's own sensitivity ("factor beta" or "factor loading") to it.

Why Multiple Factors?

Multifactor models let investors and managers **measure and hedge exposure** to specific economic risks individually — e.g., a pension fund worried specifically about inflation risk can measure and adjust its inflation-factor beta without needing to alter its overall market beta. This is far more precise risk management than a single aggregate beta permits.

> *A two-factor model return equation looks like: $R_i = E(R_i) + \beta_{i1} F1 + \beta_{i2} F2 + e_i$, where F1 and F2 are the surprises in two distinct macro factors (e.g., unexpected GDP growth and unexpected inflation), and each security has its own sensitivity (β_{i1}, β_{i2}) to each.*

LO2 · Arbitrage Pricing Theory

Stephen Ross developed the **arbitrage pricing theory (APT)** in 1976. Like the CAPM, it predicts a security market line — but reaches it via a completely different logical path, resting on three propositions: (1) returns follow a factor model, (2) enough securities exist to diversify away idiosyncratic risk, and (3) well-functioning markets do not allow arbitrage opportunities to persist.

Arbitrage vs. Risk–Return Dominance

Arbitrage is earning a riskless profit with zero net investment — e.g., if Canadian Natural Resources stock traded at C$42 on one exchange and $40 on another, buying low and selling high nets a riskless $2/share profit. The **Law of One Price** says equivalent assets must have the same price; arbitrageurs enforce it. Crucially, an arbitrage argument is **stronger** than the CAPM's risk–return dominance argument: dominance requires *many* investors each making small portfolio shifts to move prices, whereas a true arbitrage opportunity motivates any single investor to take an *unlimited* position, so it only takes a few sophisticated arbitrageurs to eliminate the mispricing almost instantly.

Well-Diversified Portfolios

In a well-diversified portfolio (many securities, no single one dominating the weights), firm-specific risk is diversified away almost entirely, leaving **only** factor (systematic) risk. This is the key structural fact the APT exploits: for such portfolios, risk is fully summarized by factor betas alone.

The No-Arbitrage Equation

Arbitrage activity forces the alpha of any well-diversified, zero-beta portfolio to zero — otherwise it would represent a riskless profit opportunity. This directly implies that for *any* well-diversified portfolio P, the risk premium must equal beta times the factor risk premium (see Formulas page, F1) — exactly the same single-factor SML the CAPM predicts, but derived here purely from the absence of arbitrage, without any of the CAPM's strong assumptions about investor behaviour.

> *The APT and CAPM reach the identical single-factor SML through very different arguments: the CAPM assumes utility-maximizing, mean-variance-optimizing investors; the APT requires only well-diversified portfolios and the absence of arbitrage. The APT's weaker assumptions are often seen as a theoretical advantage.*

LO3 · The APT, the CAPM, and the Index Model

The APT's no-arbitrage SML is guaranteed to hold **exactly** only for well-diversified portfolios ("Level 0" portfolios in the textbook's terminology). For any *individual* security or poorly diversified portfolio ("Level 1"), some residual (non-systematic) risk remains, so the SML holds only **approximately** — deviations are unbiased on average across many such securities, but the precision of the SML's prediction for any single security deteriorates as its residual risk increases.

This creates a genuine practical limitation: because true arbitrage requires strictly riskless, zero-net-investment positions, and individual mispriced stocks still carry firm-specific risk, pure risk-free arbitrage may not be fully available to correct *individual* security mispricing. Enforcement instead falls to well-diversified portfolio managers exploiting "near-arbitrage" opportunities across many such securities at once — a weaker, slower correction mechanism than pure arbitrage, but still a meaningful force pushing prices toward the SML.

In short: the APT's no-arbitrage condition guarantees the single-factor SML holds precisely for the class of well-diversified portfolios that a well-run index or mutual fund would hold — which is exactly the setting in which the Chapter 8 single-index model, and its use of alpha and residual risk, was built to operate.

LO4 · A Multifactor APT

Generalizing to multiple factors, the APT’s no-arbitrage logic still applies: a well-diversified portfolio’s total risk premium must equal the **sum** of the compensation earned for each separate systematic risk exposure — each factor beta multiplied by that factor’s own risk premium (see Formulas page, F2, and the worked example on the Examples page).

Constructing a Replicating (Factor) Portfolio

Given any well-diversified portfolio P with known factor betas, one can construct a competing portfolio Q from the pure factor portfolios (plus T-bills) that has **identical** factor betas to P. If P and Q had different expected returns despite identical risk exposures, an investor could go long the higher-expected-return portfolio and short the lower one, in equal dollar amounts — a zero-net-investment position whose factor exposures fully cancel out, leaving a riskless, positive expected profit. This is precisely how the multifactor no-arbitrage SML is enforced in practice.

An important nuance versus the single-factor case: in a multifactor world, a beta greater than 1 on any *one* factor no longer automatically signals an "aggressive" (high risk premium) portfolio the way it did under the single-index model — whether a portfolio is aggressive or defensive now depends on the *sum* of its risk-premium contributions across all priced factors, which can offset each other in ways a single beta cannot capture.

LO5 · The Fama-French (FF) Three-Factor Model

Rather than specifying macroeconomic variables directly as factors, the currently dominant empirical approach uses **firm-characteristic-based factors** — portfolio returns constructed from observable firm traits that historically predict returns better than the CAPM alone. The best-known version is the **Fama-French three-factor model** (see Formulas page, F3), which adds two factors to the market factor:

- **SMB (Small Minus Big) —** the return of a portfolio of small-capitalization stocks in excess of the return on a portfolio of large-cap stocks — a "size factor."
- **HML (High Minus Low) —** the return of a portfolio of high book-to-market stocks in excess of the return on a portfolio of low book-to-market stocks — a "value factor" (high book-to-market roughly corresponds to "value" stocks; low book-to-market to "growth" stocks).

Fama and French justify these choices on **empirical** grounds: firm size and book-to-market ratio have long been observed to predict deviations of average returns from the simple CAPM's prediction. The proposed economic rationale is that firms with high book-to-market ratios are more likely to be in financial distress and thus more sensitive to broad economic conditions — so SMB and HML may be proxying for exposure to macro risk factors not fully captured by the market portfolio alone, even though neither is an obviously "fundamental" risk factor in its own right.

A Caution on Purely Empirical Factors

A recognized weakness of firm-characteristic-based models like Fama-French: because the factors are chosen for their *historical* ability to explain returns rather than derived from theory, risk-premium estimates for characteristics like firm size have **not proven fully stable** over time since first discovered — a pattern that shows up repeatedly in empirical asset pricing (a factor "works" in the sample used to discover it, then weakens once traders crowd into strategies built around it).

Chapter 10 — Big-Picture Takeaways

- Multifactor models extend the single-index model, letting each security respond to several distinct systematic risk sources with its own factor beta for each.
- The APT reaches a single-factor SML identical to the CAPM's, but from weaker assumptions: only well-diversified portfolios and the absence of arbitrage, not full mean-variance investor optimization.
- The APT's no-arbitrage SML holds exactly for well-diversified portfolios but only approximately for individual securities, which still carry residual (non-systematic) risk.
- In a multifactor world, "aggressive vs. defensive" depends on the sum of risk-premium contributions across all priced factors, not on any single beta in isolation.
- The Fama-French three-factor model adds firm-characteristic-based SMB (size) and HML (value) factors to the market factor, chosen for their empirical explanatory power rather than a fully settled theoretical rationale.

- Purely empirical, characteristic-based factors carry a real risk: their apparent risk premiums are not guaranteed to be stable once discovered and widely traded upon.

The multifactor SML (F2) and its arbitrage-based enforcement are the most heavily tested content from this chapter.

F1 | Single-Factor No-Arbitrage SML (APT)

$$E(R_P) = \beta_P \times E(R_M) \quad \text{(for any well-diversified portfolio P)}$$

Identical in form to the CAPM's SML, but derived purely from the absence of arbitrage among well-diversified portfolios — no assumption of mean-variance-optimizing investors is required.

F2 | Multifactor (Two-Factor) SML

$$E(r_P) = r_f + \beta_{P1}[E(r_1) - r_f] + \beta_{P2}[E(r_2) - r_f]$$

r_1, r_2 = returns on the two factor portfolios (each with a beta of 1 on its own factor, 0 on the other). Total risk premium is the sum of the compensation for each separate factor exposure.

F3 | Fama-French Three-Factor Model

$$R_i = \alpha_i + \beta_{iM} \times R_M + \beta_{iSMB} \times SMB + \beta_{iHML} \times HML + e_i$$

SMB = Small-Minus-Big (size factor); HML = High-Minus-Low book-to-market (value factor). All are firm-characteristic-based portfolio returns, not macroeconomic variables directly.

F4 | Arbitrage Profit from Mispricing

$$\text{Zero-net-investment profit} = \$1 \times [E(r_Q) - E(r_A)]$$

where Q is constructed to match A's factor betas exactly

If a well-diversified portfolio A's expected return differs from a replicating portfolio Q with identical factor betas, going long \$1 in Q and short \$1 in A creates a riskless, zero-net-investment profit equal to the return gap — the arbitrage that no-arbitrage pricing rules out.

EXAMPLE 10.3 | The Multifactor SML

Given: Two factor portfolios: $E(r_1)$=10%, $E(r_2)$=12%. Risk-free rate = 4%. Portfolio A has β_A1=0.5 and β_A2=0.75.

Step 1 — Risk premium from each factor:

$$\text{Factor 1: } \beta_{A1} \times [E(r_1)-r_f] = 0.5 \times (10-4) = 3\%$$
$$\text{Factor 2: } \beta_{A2} \times [E(r_2)-r_f] = 0.75 \times (12-4) = 6\%$$

Step 2 — Total required return:

$$E(r_A) = 4 + 3 + 6 = 13\%$$

Notice: even though both of Portfolio A’s factor betas are below 1, its total risk premium (9%) can still exceed the historical market risk premium — a reminder that "aggressive vs. defensive" depends on the sum across all priced factors, not any single beta.

EXAMPLE 10.4 | Mispricing and Arbitrage

Given: Using Example 10.3’s data, suppose Portfolio A is actually priced to offer only 12% (not the fair 13%).

Step 1 — Build a replicating portfolio Q with A’s exact factor betas:

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Weights: 0.5 in factor portfolio 1, 0.75 in factor portfolio 2,
−0.25 in T-bills
E(r_Q) = 4 + 0.5(10−4) + 0.75(12−4) = 13%  (same betas as A, by
construction)
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Step 2 — Arbitrage trade: long $1 in Q, short $1 in A:

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Expected profit = $1 × 0.13 − $1 × 0.12 = $0.01, with ZERO net
investment
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The position is riskless because Q and A have identical factor exposures, so being long one and short the other cancels all systematic risk — leaving a pure, riskless $0.01 profit per dollar. This is the arbitrage opportunity that no-arbitrage pricing (LO2) rules out in true equilibrium.

PRACTICE PROBLEM | Kananaskis Capital Management

Scenario

You are a quantitative analyst at Kananaskis Capital Management in Calgary, working with a two-factor model. Factor 1 is unexpected GDP growth; Factor 2 is unexpected inflation. The risk-free rate is 3%. Factor portfolio 1 has an expected return of 9%; factor portfolio 2 has an expected return of 6%.

Portfolio	Beta on Factor 1 (GDP)	Beta on Factor 2 (Inflation)
Bow River Fund	1.2	−0.4
Spray Valley Fund	0.6	0.9

Part A — Multifactor SML

(i) Calculate the fair (no-arbitrage) expected return for the Bow River Fund.

(ii) Calculate the fair expected return for the Spray Valley Fund.

(iii) The Bow River Fund has a negative beta on the inflation factor. Explain in plain language what this means for its risk premium contribution from that factor.

Part B — Detecting an Arbitrage Opportunity

Suppose the Bow River Fund is actually priced in the market to offer an expected return of 14.5% (not the fair value from Part A).

(i) Is there an arbitrage opportunity? If so, is Bow River overpriced or underpriced relative to its fair value?

(ii) Describe the weights of a replicating portfolio Q built from the two factor portfolios and T-bills that matches Bow River's exact factor betas.

(iii) Describe the specific long/short position an arbitrageur would take, and calculate the riskless profit per $1 invested.

Part C — Aggressive vs. Defensive in a Multifactor World

(i) Using only the market-factor beta convention from Chapter 8 (beta > 1 = aggressive), would you call the Bow River Fund "aggressive" or "defensive"? Explain why this single-factor label can be misleading here.

(ii) Compare the total risk premiums of Bow River and Spray Valley (from Part A). Which fund actually carries the larger total systematic risk premium, and does that match what a naive single-beta view would suggest?

Part D — Fama-French Application

Kananaskis is considering adding a Fama-French three-factor analysis to its process.

(i) Define SMB and HML in your own words, using the exact definitions from LO5.

(ii) A junior colleague argues that HML is unambiguously a "fundamental" risk factor because it reliably predicts returns historically. Evaluate this claim, referencing the chapter’s caution about purely empirical factors.

Hints & Guidance

Part A tests LO4 and Formula F2 (mirror Example 10.3 exactly). Part B tests LO2 and Formula F4 (mirror Example 10.4 — remember the replicating portfolio must match BOTH betas). Part C synthesizes LO1 and LO4. Part D tests LO5 directly.